Name: INSTRUCTOR SOLUTIONS **Set:** _____

Student Number: _____

British Columbia Institute of Technology
Burnaby, B.C.

MSYS 3386 Heating Systems

Instructor: Dr. Joseph Cheung, P.Eng.

Mid-Term Examination

(Time allowed: 110 minutes)

Please find attached data sheets and formulae sheet at the back of the exam. All calculations shall be rounded off to 3 significant figures.

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Total Marks:		/100

Date: October 27, 2017

INSTRUCTOR SOLUTIONS

INSTRUCTOR SOLUTIONS

British Columbia Institute of Technology
Mechanical Engineering Technology (Systems Option)
MSYS 3386 Heating Systems

SET: _____

Name: _____

1. A building has a heating load of 2.1 million BTU/h and three condensing boilers of 95% efficient is selected for this building. Each boiler has a heating output of 790,000 BTU/h and they are located in the mechanical room as shown in Figure 1.

   a. Determine the size of the labeled gas pipes. (5 marks)

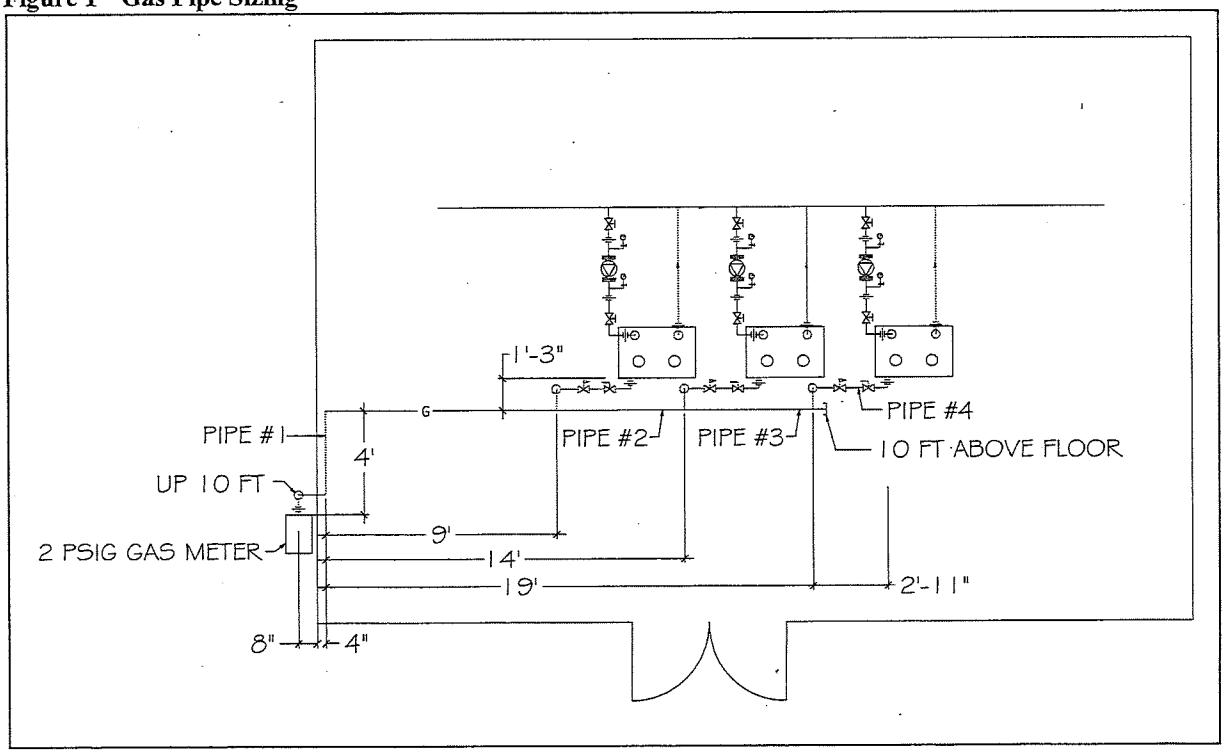
Description	Pipe Sizes	Gas Load (BTU/h)
Pipe #1	1 1/4"	2495,000
Pipe #2	1"	1,663,000
Pipe #3	3/4"	832,000
Pipe #4	1 1/4"	832,000

TOTAL RUN OF PIPE IS 50 FT.

   b. Discuss the benefits of designing the system with three equal size boilers. (3 marks)

- REDUNDANCY FOR EQUIPMENT TO ROUTINE MAINTENANCE AND MECHANICAL FAILURE
- REDUNDANCY PROVIDED DURING NON PEAK HEATING SEASON
- PROVIDE INCREASE IN OPERATING EFFICIENCY WITH MORE TURN DOWN RATIO to MEET THE PART LOAD EFFICIENCY.

Figure 1 Gas Pipe Sizing



INSTRUCTOR SOLUTION

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2. Describe the construction of oil fuel storage tank. (4 marks)
- DOUBLE WALL FIBREGLASS OR STEEL CONSTRUCTION
- PRUVIDE TRAP FOR LEAKING OIL AND TO PREVENT THE SURROUNDING SOIL BEING CONTAMINATED
- CAVITY BETWEEN WALLS MUST BE MONITORED WITH LEAKS
- MUST BE CORROSION RESISTANCE
- STORAGE EQUAL TO THREE TO FOUR WEEKS OF PEAK CONSUMPTION

3. What would the minimum requirements be for terminating boiler vents on a roof? (2 marks)
- MINIMUM 2FT ABOVE THE HIGHEST POINT
- 10 FT FROM ANY ADJACENT VERTICAL WALLS

4. Describe the benefits of using dual burners in large central heating plant (2 marks)
- ALLOW FUEL TO BE SWITCHED BETWEEN GAS AND OIL
- PROVIDE BACK-UP FOR EMERGENCY

5. What would the temperature be for the flue gas to start condensing for natural gas fired condensing boilers? (1 mark)
130F

6. When would the steam boiler be preferred over the hot water boiler? (1 mark)
WHEN STEAM IS ALSO USED FOR OTHER PROCESSES AS WELL AS FOR HEATING.

7. At what heating capacity would residential heating boilers be tested for Annual Fuel utilization Efficiency (AFUE)? (1 mark)
< 300,000 BTH/H.

8. Describe part load conditions for heating systems? (2 marks)
PART LOAD IS THE PERCENTAGE OF THE LOAD DURING HEATING SEASON AS COMPARED WITH THE PEAK LOAD CONDITION.

9. What would the reference be used to design the vents for natrual draft boilers? (1 mark)
GAS SAFETY CODE B-149

10. What would the reference be used to design the vents for condensing boilers? (1 mark)

MANUFACTURER INSTALLATION MANUAL / GUIDE

11. What are the three different ways to define boiler efficiencies? (3 marks)

- COMBUSTION - INPUT MINUS STACK LOSS DIVIDED BY INPUT
- OVERALL EFFICIENCY - GROSS OUTPUT / INPUT
- SEASONAL EFFICIENCY - OPERATION DURING PART LOAD

12. What should the condensate from a condensing boiler be handled before discharging to the building drainage system? (1 mark)

CONDENSATE SHOULD BE TREATED WITH ACID NEUTRALIZER BEFORE DISCHARGING TO THE DRAIN

13. What are the considerations to select electric boilers for heating purposes? (1 mark)

VENTING IS NOT REQUIRED.
ALTERNATIVE FOR RESTRICTIVE EMISSION REGULATION.

14. Discuss the operating and physical characteristics of heating boilers to demonstrate the differences between high mass and low mass boilers? (6 marks)

HIGH MASS BOILER - MORE WATER CONTENT AND HEAVIER COMBUSTION CHAMBER.
- FIRE TUBE BOILER, STANDING HEAT LOSS COULD BE CONSIDERED AS INEFFICIENCY. ALLOWS FOR LOAD SWING WHERE LARGE AMOUNT OF HEAT IS REQUIRED IN SHORT PERIOD

LOW MASS BOILER - MINIMAL WATER CONTENT
- WATER TUBE BOILER
  - PROVIDE FASTER RESPONSE TO LOAD FLUCTUATION
  - SENSITIVE TO HEATING DEMAND AND THE VOLUME FLOW RATE.

15. What are the safety devices applicable to the boiler ? (2 marks)

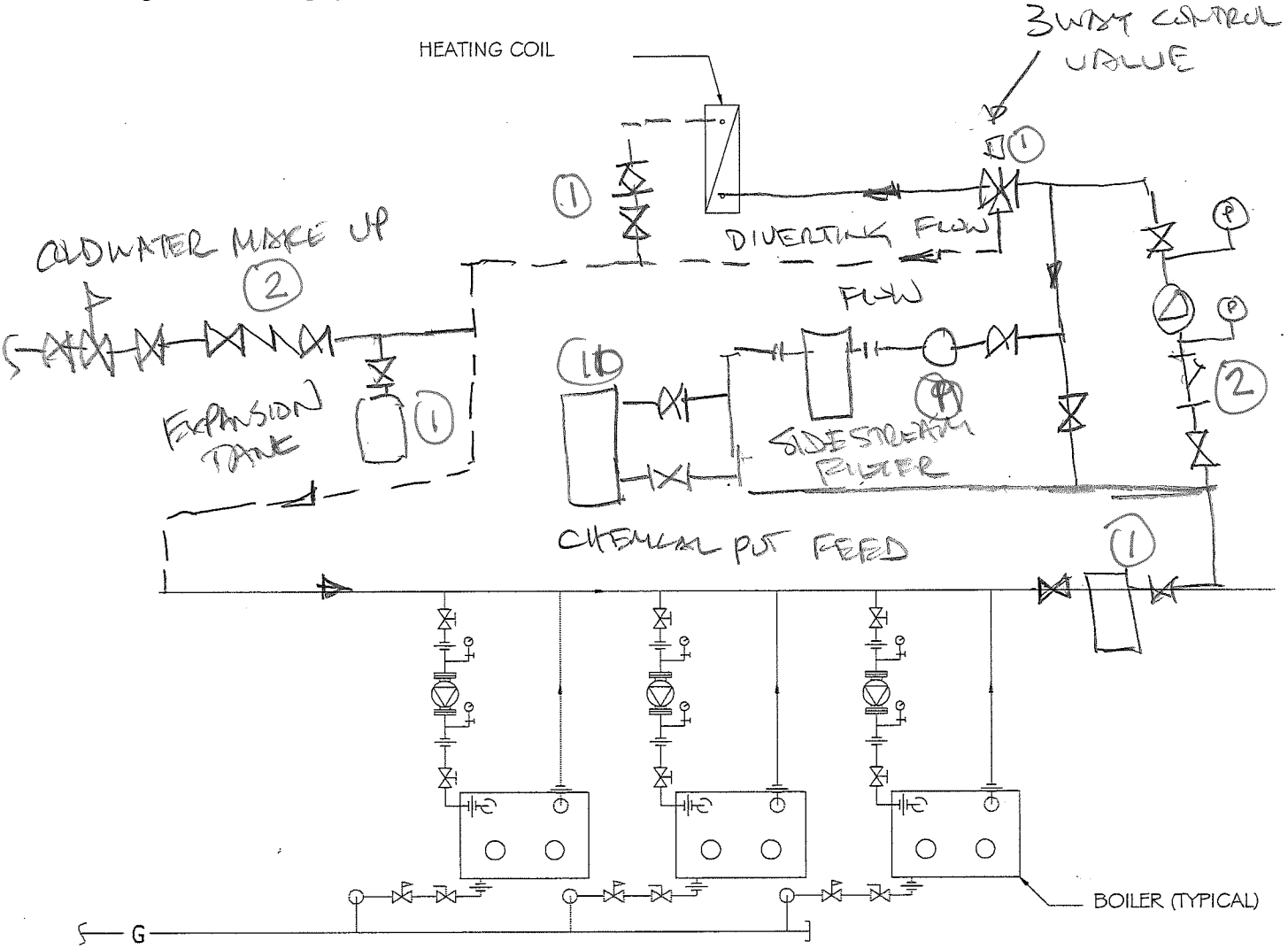
TEMPERATURE AND PRESSURE RELIEF VALVES
LOW WATER CUT OFF SWITCH.

SET: _____

Name: _____

16. Complete the heating schematic as shown in Figure 2. The heating coil shall be connected to a 3-way control valve to allow water be diverted to maintain a constant flow system (10 marks)

Figure 2 Heating System Schematic



17. What is the type of material used in older type of boiler that poses health risk? (1 mark)

ASBESTOS

18. Discuss the deficiency of designing hyronic heating systems to provide the heating in buildings and how to address the deficiency of the design. (2 marks)

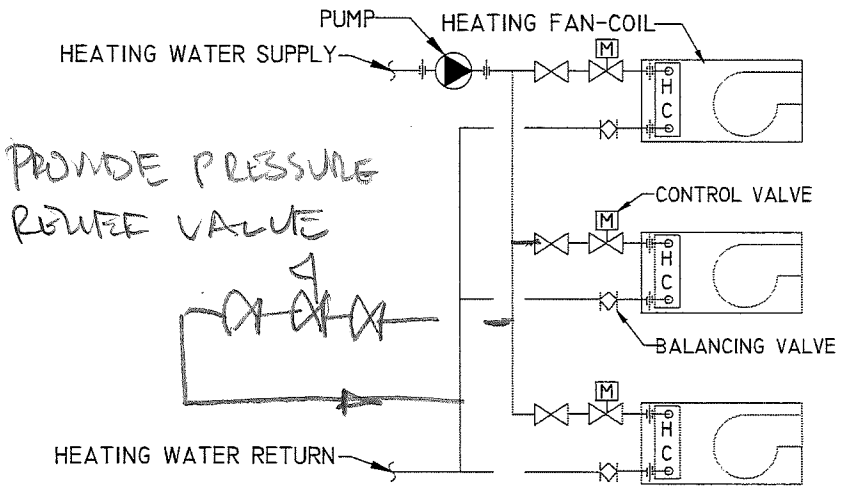
- NO VENTILATION PROVIDED - SEPARATE VENTILATION SYSTEM TO MAINTAIN INDOOR AIR QUALITY
- ADD A DEDICATED OUTDOOR AIR SYSTEM WITH A HEAT RECOVERY UNIT TO DELIVER 100% TEMPERED OUTDOOR AIR TO THE SPACE.

SET: _____

Name: _____

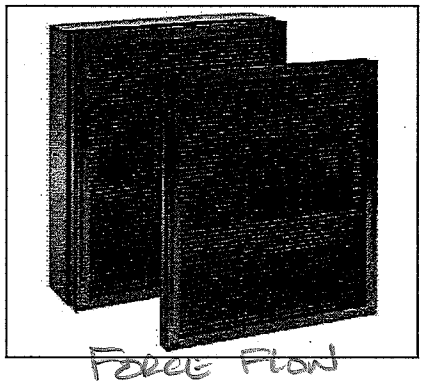
19. The heating water pump is operating at a constant speed. Describe what the problem and also the provision needed be to protect the pump from damage when the two way control valves are starting to close. (2 marks)

Figure 3 Heating Water System

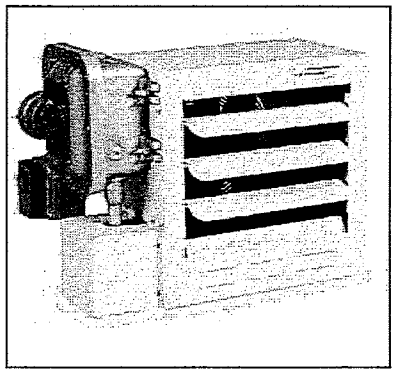


PRESSURE INCREASES AND PUMP WILL FAIL
ADD PRESSURE RELIEF VALVE TO RELIEF PRESSURE BUILD-UP.

20. What are the names for the heaters as shown below? (2 marks)



Name: CONVECTOR



Name: UNIT HEATER

21. What is the difference between a 2-pipe and a 4-pipe fan coil? (1 mark)

2-PIPE FOR EITHER HEATING OR COOLING
4-PIPE HAS BOTH HEATING AND COOLING COIL

22. What is the characteristic of propane gas at a temperature lower than 32F? (1 mark)

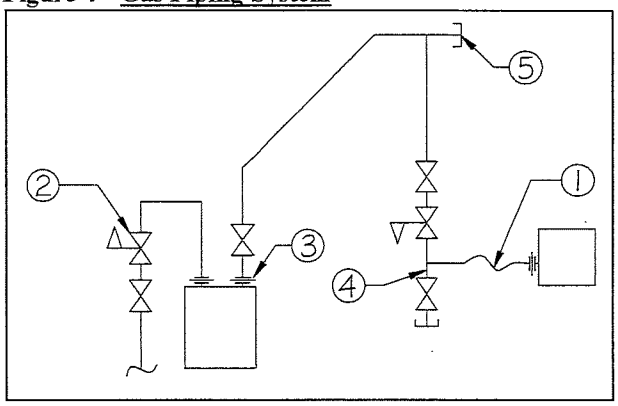
WILL NOT GASIFY AT TEMPERATURES LOWER THAN 32F

INSTRUCTOR SOLUTIONS

23. Name the noted symbols as shown in Figure 4. (5 marks)

1	FLEXIBLE CONNECTOR
2	SERVICE REGULATOR
3	UNION
4	DRIP LEG
5	END CAP / PIPING EXTENSION

Figure 4 Gas Piping System



24. What are the environmental impacts on burning fossil fuels? (2 marks)

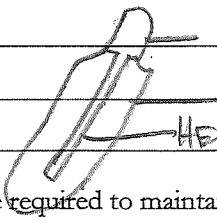
PRODUCE SULPHUR DIOXIDE SO_2 + NITROGEN OXIDE AND CAUSE THE FORMATION OF SMOG.

25. List four different design references for energy efficiency designs? (4 marks)

PART 10 OF 2012 BCBC
2015 NATIONAL ENERGY CODE OF CANADA
ASHRAE STANDARD 90.1 ENERGY STD FOR BUILDING
LEED - LEADERSHIP IN ENVIRONMENTAL AND ENERGY DESIGN.

26. Describe the principal design of an evacuated tube collector. (2 marks)

HEAT PIPE ENCASED IN AN EVACUATED TUBE



27. What would be required to maintain the operation of closed loop solar thermal panels in winter? (1 mark)

ADD GLYCOL TO THE WATER LOOP TO LOWER THE FREEZING POINT OF WATER

28. Who is typically responsible to provide sizing of gas pressure regulators? (1 mark)

GAS JOURNEYMAN.

29. What is the separation distance between the termination point of a vent and a window opening? (1 mark)

10 ft

30. What is the separation distance between the termination point of a vent and an appliance vent outlet? (1 mark)

3 ft

31. What is the maximum diameter of copper pipes permitted for gas piping? (1 mark)

1 1/4"

32. What are the two types of joints permitted for gas copper pipes? (2 marks)

SOLDER JOINT OR FLARE JOINT
BRAZED

33. Gas pipe must be pressure tested after it is installed. How long should the test pressure be held for the pressure test? (1 mark)

10 MINUTES WHILE PRESSURE TESTING GAS PIPES

34. A 1" gas pipe is sized to run a 100 ft of gas pipe with a gas pressure of 2 psig for a gas load of 1632 ft³ of gas in Vancouver at sea level (i.e. 14.62 psia). What would the pipe size be if the same gas pipe is installed in an location where the barometric pressure is 12.75 psia? (4 marks)

$P_1V_1 = P_2V_2$. $P_1 = 14.62 + 2 = 16.62$ psi

$V_1 = 1632\ ft^3$; $P_2 = 12.75 + 2 = 14.75$ psi

$V_2 = \frac{16.62 \times 1632}{14.75} = 1838.4\ ft^3$ 1 1/4"

35. What is the maximum run of 3/8" gas pipes connecting to gas appliance with input less than 15,000 BTU/h? (1 mark)

25 ft

36. What is the maximum size of gas pipes that is allowed to have threaded connections? (1 mark)

2"

37. What is the maximum gas pressure allowed in a 1 or 2 family dwellings and row housing? (1 mark)

2 PSIG

38. What is the maximum gas pressure allowed in commercial buildings? (1 mark)

5 PSIG

39. What is the type of boiler used thermoplastic or CPVC pipe for venting? (1 mark).

CONDENSING BOILER

SET: _____

Name: _____

40. What is the benefit to install all electric system in a strata title residential building? (1 mark).

SEPARATE METERING OF ELECTRICAL USES

41. Describe the responsibility of mechanical designer when a building is designed for electric heating using (a) electric baseboard heaters and (b) fan-coil with electric resistance heater. (4 marks).

a) BUILDING HEAT LOSS CALCULATION

b) HEAT LOSS CALCULATION AND SPECIFICATION FOR THE HEATERS

42. What are the typical voltages used for low voltage controls of mechanical equipment? (1 mark).

24V OR 12V

43. List the pilot device used to schedule the operation of mechanical equipment. (1 mark).

TIME CLOCK

44. List the pilot device used to control building space temperature. (1 mark).

THERMOSTAT

45. What would the heating output of a 208V/1Ø 12 KW electric heater be if the heater is supplied with 115V/1Ø electrical power? (1 mark).

POWER = $VI = V^2/R$ $I = \frac{V}{R}$ $V = IR$

$12{,}000 = 208^2/R$ $\therefore R = 3.605\ \Omega$

POWER = $115^2/3.605 =$ **3.7 KW**

46. List the two different types of electric duct heaters. (2 marks).

SHEATH TUBULAR DUCT HEATER / OPEN COIL DUCT HEATER

47. What applications would sheath tubular duct heater be recommend? (2 marks).

AREAS WITH DUST LADEN ATMOSPHERE AND DOWNSTREAM OF A HUMIDIFYING COIL

48. What is the benefit of heating outdoor air with electric heating resistance coil verse hydronic heating coil? (1 mark).

POTENTIAL FREEZING UP OF WATER COIL

49. What would the safety device be for the control of electric resistance heating coil? (1 mark).

HIGH LIMIT TEMPERATURE SENSOR

INSTRUCTOR SOLUTION

Table A.3
Maximum capacity of natural gas in thousands of Btuh for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 2 psig based on a pressure drop of 1 psig

(See Clauses 6.3.4, 6.3.5, A.2.3, A.2.4, A.2.6, A.3.5, and E.1.3.)

(a) Imperial

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	1 440	3 012	5 673	11 648	17 452	33 610	53 570	94 701	193 161
20	990	2 070	3 899	8 005	11 994	23 100	36 818	65 088	132 758
30	795	1 662	3 131	6 429	9 632	18 550	29 566	52 268	106 610
40	680	1 423	2 680	5 502	8 244	15 877	25 305	44 734	91 244
50	603	1 261	2 375	4 876	7 306	14 071	22 427	39 647	80 868
60	546	1 142	2 152	4 418	6 620	12 749	20 321	35 923	73 272
70	503	1 051	1 980	4 065	6 090	11 729	18 695	33 049	67 409
80	468	978	1 842	3 782	5 666	10 912	17 392	30 746	62 711
90	439	917	1 728	3 548	5 316	10 238	16 318	28 848	58 840
100	414	867	1 632	3 351	5 022	9 671	15 414	27 249	55 580
125	367	768	1 447	2 970	4 451	8 571	13 661	24 151	49 260
150	333	696	1 311	2 691	4 032	7 766	12 378	21 882	44 633
175	306	640	1 206	2 476	3 710	7 145	11 388	20 131	41 062
200	285	596	1 122	2 303	3 451	6 647	10 594	18 728	38 200
250	252	528	994	2 042	3 059	5 891	9 389	16 599	33 856
300	229	478	901	1 850	2 772	5 338	8 507	15 040	30 676
350	210	440	829	1 702	2 550	4 911	7 827	13 836	28 221
400	196	409	771	1 583	2 372	4 568	7 281	12 872	26 255
450	184	384	724	1 485	2 226	4 286	6 832	12 077	24 634
500	174	363	683	1 403	2 102	4 049	6 453	11 408	23 269
600	157	329	619	1 271	1 905	3 669	5 847	10 337	21 083
700	145	302	570	1 170	1 752	3 375	5 379	9 510	19 396
800	135	281	530	1 088	1 630	3 140	5 004	8 847	18 045
900	126	264	497	1 021	1 530	2 946	4 695	8 301	16 931
1 000	119	249	470	964	1 445	2 783	4 435	7 841	15 993
1 200	108	226	426	874	1 309	2 521	4 019	7 104	14 490
1 400	99	208	392	804	1 204	2 320	3 697	6 536	13 331
1 600	92	193	364	748	1 120	2 158	3 439	6 080	12 402
1 800	87	181	342	702	1 051	2 025	3 227	5 705	11 636
2 000	82	171	323	663	993	1 913	3 048	5 389	10 992

(Continued)

Table A.1
Maximum capacity of natural gas in thousands of Btuh for Schedule 40 pipe and plastic pipe, including fittings, for pressures of less than 7 in w.c. based on a pressure drop of 0.5 in w.c.

(See Clauses 6.3.2, 6.3.5, A.2.3, A.2.4, A.2.6, and A.3.5.)

(a) Imperial

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	156	326	614	1 261	1 890	3 639	5 800	10 253	20 914
20	107	224	422	867	1 299	2 501	3 986	7 047	14 374
30	86	180	339	696	1 043	2 008	3 201	5 659	11 543
40	74	154	290	596	893	1 719	2 740	4 843	9 879
50	65	137	257	528	791	1 524	2 428	4 293	8 756
60	59	124	233	478	717	1 380	2 200	3 889	7 933
70	54	114	214	440	659	1 270	2 024	3 578	7 299
80	51	106	199	409	613	1 181	1 883	3 329	6 790
90	48	99	187	384	576	1 109	1 767	3 123	6 371
100	45	94	177	363	544	1 047	1 669	2 950	6 018
125	40	83	157	322	482	928	1 479	2 615	5 333
150	36	75	142	291	437	841	1 340	2 369	4 832
175	33	69	131	268	402	774	1 233	2 180	4 446
200	31	64	121	249	374	720	1 147	2 028	4 136
250	27	57	108	221	331	638	1 017	1 797	3 666
300	25	52	98	200	300	578	921	1 628	3 321
350	23	48	90	184	276	532	847	1 498	3 056
400	21	44	83	171	257	495	788	1 394	2 843
450	20	42	78	161	241	464	740	1 308	2 667
500	19	39	74	152	228	438	699	1 235	2 519
600	17	36	67	138	206	397	633	1 119	2 283
700	16	33	62	127	190	365	582	1 030	2 100
800	15	30	57	118	177	340	542	958	1 954
900	14	29	54	111	166	319	508	899	1 833
1 000	13	27	51	104	156	301	480	849	1 732
1 200	12	24	46	95	142	273	435	769	1 569
1 400	11	23	42	87	130	251	400	708	1 443
1 600	10	21	39	81	121	234	372	658	1 343
1 800	9	20	37	76	114	219	349	618	1 260
2 000	9	19	35	72	108	207	330	583	1 190

(Continued)